

# GURURAGHURAMAN SETHURAMAN

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## Executive Summary

Embedded Systems & Electrical Engineer specializing in EV powertrain technologies, motor fault diagnostics, and database architecture for engineering data systems. Author of 2 IEEE publications on IPMSM demagnetization fault detection combining Finite Element Analysis (FEA) with machine learning. Hands-on experience across embedded firmware (Embedded C), CAN bus validation on live high-voltage (HV) testbeds, and relational database design, gained through roles at Robert Bosch, Schneider Electric, and AInsurCo. Targeting Embedded Systems, Automotive Electronics, Motor Control, or EV Powertrain Engineering roles.

## Professional Experience

### AInsurCo

Aug 2025 – Present

#### Database Architect & Python Developer - Remote

London, The United Kingdom

- Designed and maintained 10+ relational database schemas/tables supporting core insurance data operations, applying normalization and indexing best practices.
- Wrote and optimized 30+ SQL queries and stored procedures for reporting, validation, and data-retrieval workflows across the platform.
- Resolved 15+ data-integrity and schema-consistency issues through constraint enforcement and automated validation checks.
- Built 5+ Python automation scripts to run scheduled batch jobs (30% of role scope, with database architecture as primary focus) supporting data processing tasks on the database layer.

### Robert BOSCH Limited

June 2024 – July 2024

#### Embedded Hardware Intern - On-site

Bangalore, India

- Developed and validated 5+ Embedded C firmware modules implementing CAN communication layers for EV benchmarking infrastructure and VCU interfaces.
- Analyzed 100+ CAN frames using CANalyzer, performing frame-level logging, DLC verification, error-state capture, and fault-injection analysis across 3+ live HV testbeds.
- Contributed to technical evaluation of E-axle architecture across 2 HV subsystem configurations, documenting integration behavior per internal test protocols.

### Schneider Electric

July 2023 – July 2023

#### Supply Chain Intern (Electronics Division) - On-site

Chennai, India

- Validated 150+ PCB components across power-electronics manufacturing lines, identifying non-conformances against design specifications.
- Analyzed component lifecycle and supply dependencies for 8+ critical converter subsystems, supporting procurement risk assessment.

### Igarashi Motor Pvt Ltd

Feb. 2023 – Feb. 2023

#### Manufacturing and Quality Intern - On-site

Chennai, India

- Audited motor assembly workflows across 3+ production stations, identifying redundant checkpoints and contributing to an estimated 10–15% reduction in inspection cycle time.
- Reviewed and improved BOM accuracy for 20+ line items, strengthening defect-traceability documentation for small-motor production.

### Sri Balaji Assembles and Plastics India Pvt Ltd

July 2022 – Aug. 2022

#### Quality Intern (Electronics Assembly) - On-site

Chennai, India

- Supported IPC/ISO-compliant quality control checks across 50+ electronic sub-assembly units.
- Collaborated with production teams to isolate component-level defects across 10+ assembly batches, reducing rework cycles.

## Education

### Grenoble-INP PHELMA

Jan. 2025 – June 2025

Erasmus Exchange Program in Nanotechnology

Grenoble, France

- Coursework: Advanced CMOS, Microsystems, Advanced VLSI, Spintronics, Semiconductor Physics, Nanoelectronics

### Amrita Vishwa Vidyapeetham

Apr. 2021 – May 2025

Bachelor of Technology in Electrical and Electronics Engineering

Coimbatore, India

- Coursework: Embedded Systems, Microprocessors & Microcontrollers, Power Electronics, Control Systems, Electric Drives, Simulation & Modeling

Publications

- Demagnetization in V-shaped IPMSM: A Fusion of Finite Element Analysis and Machine Learning  
*IEEE CONECCT, June 2025*
- Machine Learning-based Analysis on Non-uniform Demagnetization Fault in IPMSM Using FEM  
*ICPERS, May 2025*

Projects

- Demagnetization Fault Prediction in IPMSM using ML | *ANSYS Motor-CAD, Python* April 2025
- Trained and evaluated ML models across 10+ fault-severity classes using FEM-simulated datasets for early-stage detection of permanent-magnet demagnetization in EV-class IPMSM units, enabling predictive fault-monitoring for electric drivetrains.
- SOH Prediction for LiFePO4 Batteries using ML | *Raspberry Pi 4, Python, MS Excel* June 2024
- Built a regression pipeline trained on battery cycling data collected across 50+ charge/discharge cycles to predict State of Health, providing lifecycle decision support for EV battery management systems.
- Comparative Study of Subsystems in 32-bit RISC Processor | *Vivado* Jan. 2025
- Designed and implemented 3 core subsystems (ALU, data memory, register file) in Verilog HDL, synthesized and tested across multiple FPGA configurations for timing and resource-utilization comparison.
- MTJ-based Random Number Generator | *Python* May 2025
- Developed and tested a stochastic solver for Magnetic Tunnel Junction (MTJ)-based random number generators across 25+ simulation trials for low-power embedded applications.

Technical Skills

-  Languages: C, Embedded C, Verilog, Python, SQL
-  Database & Data Systems: Relational Database Design, Schema Architecture, Query Optimization, Batch Processing, Data Integrity
-  Microcontrollers & Platforms: STM32, dsPIC, ESP32, Arduino, Raspberry Pi
-  Protocols: CAN, UART, SPI, I2C
-  Control Systems & Power Electronics: Motor Control, Inverter Subsystems, HV Architecture, IPMSM Diagnostics
-  Automotive & EV Tools: CANalyzer, CANoe (basic), EV Benchmarking, VCU Interface Testing
-  EDA & Simulation: MATLAB/Simulink, ANSYS Motor-CAD, Vivado, Multisim, ETAP, AutoCAD
-  Software/Tools: Git, Docker, CI/CD, MPLAB X

Languages

English- C1 (IELTS - 7.0)    Deutsch- A1 (GOETHE)    Tamil- C2 (Native)    Telugu- C2 (Native)